

SmartBugs - Previous Test Execution Report

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1 Introduction

This document reports the execution results of the pre-modification test cases defined in the *Previous Test Case Specification* document.

The goal of this document is to establish a testing baseline for the original SmartBugs system before applying the proposed maintenance and evolution activities.

The executed tests focus on the original SmartBugs workflow, including file collection, tool loading, task generation, Solidity compiler handling, and main orchestration logic.

2 Test Environment

The tests were executed on the original version of the SmartBugs system, before introducing the proposed changes.

Item	Value
System under test	Original SmartBugs system
Operating system	Windows
Python version	Python 3.11.0
Testing framework	pytest 9.0.3
Pytest plugins	pytest-cov 7.1.0, pytest-mock 3.15.1, pytest-timeout 2.4.0
Executed test file	<code>tests/unit/test_smartbugs.py</code>
Executed command	<code>pytest tests/unit/test_smartbugs.py</code>
Execution date	April 2026

Table 1: Pre-modification test environment

3 Execution Summary

The pre-modification test case specification defines 11 functional test cases. These test cases were validated by executing the existing SmartBugs pytest suite contained in `tests/unit/test_smartbugs.py`. The executed pytest suite contains 34 implementation-level tests, which provide evidence for the functional scenarios described in the specification.

The test execution completed successfully from a functional point of view. All collected tests in `tests/unit/test_smartbugs.py` passed.

However, the command returned a final failure status because the project configuration enforces a global coverage threshold of 70%. Since only the `test_smartbugs.py` unit test file was executed, the measured total coverage was 41.75%, below the configured threshold.

This coverage failure is not considered a failure of the specified pre-modification test cases, since no functional test assertion failed.

Item	Result
Collected tests	34
Passed tests	34
Failed tests	0
Skipped tests	0
Measured coverage	41.75%
Required coverage threshold	70%
Functional test result	Passed
Final command status	Failed due to coverage threshold

Table 2: Summary of pre-modification test execution

4 Execution Results

The following table maps the pre-modification functional test cases to the corresponding pytest tests executed in the original SmartBugs test suite.

Test Case ID	Description	Result	Evidence
TC-PRE-01	Collection of a single valid Solidity file	PASS	test_collect_single_sol_file
TC-PRE-02	Collection of multiple Solidity files using a glob pattern	PASS	test_collect_multiple_sol_files_with_glob
TC-PRE-03	Collection and task generation for a bytecode file	PASS	test_collect_hex_files, test_collect_tasks_hex_bytecode
TC-PRE-04	Task generation for runtime bytecode	PASS	test_collect_tasks_runtime_bytecode
TC-PRE-05	Basic task generation for a Solidity file and a compatible tool	PASS	test_collect_tasks_basic
TC-PRE-06	Task generation for one Solidity file and multiple tools	PASS	test_collect_tasks_multiple_tools
TC-PRE-07	Missing pragma with a solc-dependent tool	PASS	test_collect_tasks_no_pragma
TC-PRE-08	Incompatible file and tool mode	PASS	test_collect_tasks_mode_mismatch
TC-PRE-09	Loading a tool alias	PASS	test_load_tool_alias
TC-PRE-10	Handling result directory collisions	PASS	test_collect_tasks_result_dir_collisions
TC-PRE-11	Main SmartBugs orchestration workflow	PASS	test_main_orchestration_flow, test_main_freezes_settings

Table 3: Execution results of pre-modification test cases

5 Detailed Output Summary

The executed command produced the following relevant output:

```
platform win32 -- Python 3.11.0, pytest-9.0.3
plugins: cov-7.1.0, mock-3.15.1, timeout-2.4.0
collected 34 items

tests\unit\test_smartbugs.py .....

34 passed in 1.34s
Coverage failure: total of 41.75 is less than fail-under=70.00
```

The coverage failure is related to the global coverage configuration of the project and not to the failure of any test case.

6 Conclusion

The execution of the pre-modification test cases confirms that the original SmartBugs functionality covered by `tests/unit/test_smartbugs.py` behaves correctly before the proposed changes.

All functional test cases derived from the original workflow passed successfully. These results will be used as a baseline for regression testing after the implementation of the proposed modifications.